

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. The following sequences can be shown to be telescoping. Find s_n for each series and use this to determine the sum of the series.

a) $\sum_{k=1}^{\infty} \left[\cos\left(\frac{1}{k}\right) - \cos\left(\frac{1}{k+1}\right) \right]$

b) $\sum_{k=1}^{\infty} \frac{3}{k^2 + 3k}$

2. Determine if the following series are geometric or not. If it is geometric, then find a & r and determine if the series converges. If it converges, find its sum.

a) $\sum_{n=1}^{\infty} 2(3)^n$

b) $\sum_{n=0}^{\infty} 2(3)^n$

c) $\sum_{n=1}^{\infty} n\left(-\frac{1}{2}\right)^n$

d) $\sum_{n=1}^{\infty} \frac{e^n}{3^{n-1}}$

3. Determine the sum of the following series $\sum_{n=0}^{\infty} (2^{n+1} - 1) \left(\frac{1}{9}\right)^n$.

4. Create an infinite series which converges to the value $\frac{\pi}{2}$.

5. When I was a teenager, a friend and I stole soda from my parents. I took the soda bottle, drank half of what's there, and then passed it to my friend. My friend drank half of what was left, and passed it back to me. I again drank half of what was left, and passed it back to him. We kept up with this until the entire soda bottle was empty. Approximately what percentage of the soda did I drink?

6. Later in this course we will show that $\sum_{n=0}^{\infty} \frac{1}{n!} = e$. Use this fact to show that for every $a > 2$ it must be the case that $\sum_{n=0}^{\infty} \frac{1}{a^n} < \sum_{n=0}^{\infty} \frac{1}{n!}$.

We see that $\sum_{n=0}^{\infty} \frac{1}{a^n} = \sum_{n=0}^{\infty} \left(\frac{1}{a}\right)^n$ is a geometric series and if $a > 2$ then $|r| < 1$ and this series converges.

Specifically,
$$\sum_{n=0}^{\infty} \left(\frac{1}{a}\right)^n = \frac{1}{1 - \left(\frac{1}{a}\right)} = \frac{1}{\left(\frac{a-1}{a}\right)} = \frac{a}{a-1} = \frac{a-1+1}{a-1} = 1 + \frac{1}{a-1}.$$

Since $a > 2$ it must be the case that $1 + \frac{1}{a-1} < 2 < e = \sum_{n=0}^{\infty} \frac{1}{n!}$ series sums to a value less than 2.

7. Consider the following expression

$$\sqrt{2\sqrt{4\sqrt{8\sqrt{16\ldots}}}}$$

a) Find a way to rewrite this expression so it can be expressed as 2^x , where x represents some expression.

b) Take your value of x from part (a) and subtract from it $\frac{x}{2}$. Use this result to help determine the exact value of x and the sum.

c) Use your answer from part (b) to establish the value of the given expression

8. If m is a positive integer such that $\sum_{n=1}^{\infty} \log \left(1 - \frac{1}{(n+5)^2} \right) = \log \left(\frac{m}{m+1} \right)$, then what is the value of m ?

(Hint: Start by trying to algebraically rewrite the expression inside the logarithm and see if you can rewrite the series as a telescoping series)